

Flood Risk Assessment and Drainage Strategy

Millbrook Solar & Storage – Aqueduct Water Consulting

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1. Summary

Aqueduct Water Consulting prepared this Flood Risk Assessment (FRA) and outline drainage strategy. The solar arrays, BESS and substation are located wholly within Flood Zone 1 (low probability). A corridor of up to 40 m along Mill Brook lies in Flood Zone 2/3 and has been kept free of infrastructure; only the existing field access crossing is used.

2. Sources of flooding

Source	Risk on Site	Comment
Fluvial (Mill Brook)	Low	All infrastructure outside Flood Zones 2 and 3
Surface water	Low – medium	Two shallow overland flow paths in fields 4 and 9
Groundwater	Low	Clay soils, no recorded groundwater flooding
Sewers	Very low	No public sewers on Site
Reservoirs	None	Not within any reservoir flood extent

3. Drainage strategy

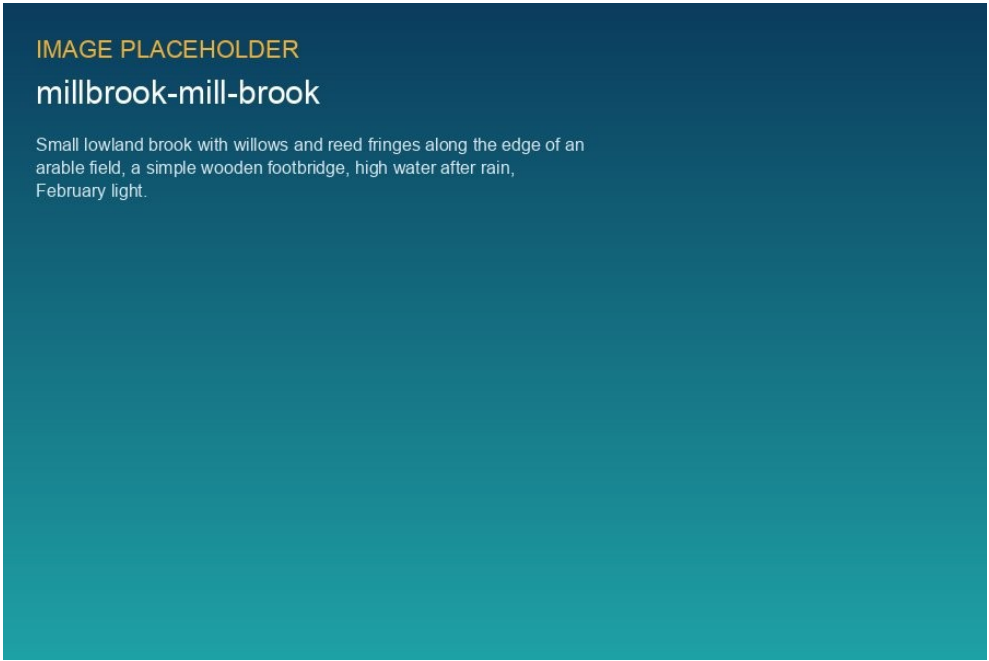
Replacing arable land with permanent grassland increases infiltration and reduces run-off. Panels do not create a significant impermeable area because rain falls off the panel edge onto vegetated ground. Impermeable areas (BESS compound 0.9 ha, substation 0.4 ha, inverter bases) drain to two attenuation basins sized for the 1-in-100-year event plus 40 % climate change allowance, with discharge limited to the greenfield rate of 2.1 l/s/ha. Swales along internal tracks follow the overland flow paths.

4. BESS fire water

Fire water run-off from the BESS compound would be contained by a lined, valve-controlled attenuation basin with 228 m³ capacity, which can be isolated and tankered away. This follows the fire and rescue service's guidance for battery storage sites.

5. Conclusion

The development is safe from flooding for its lifetime, will not increase flood risk elsewhere and provides a modest reduction in run-off rates from the Site.



Mill Brook along the western boundary, February 2026